

Traffic Flow Generation

Research Question

How can we learn a **controllable**, **realistic**, and **executable** traffic flow generator when traffic supervision is incomplete, e.g., sparse observations and missing labels?

$$\mathcal{G}_\theta : (\mathcal{R}, \mathcal{P}, \mathbf{c}) \rightarrow \mathcal{F}_{\text{sim}},$$

Input

- $\mathcal{R}$: road network structure
- $\mathcal{P}$: POIs and urban context
- $\mathbf{c}$: time and scenario condition

Output

- $\mathcal{F}_{\text{sim}}$: executable traffic flow files for simulators
- Contains flows indexed by intersection i , road link e , movement m , and time t

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Why It Matters

Signal control algorithms need realistic, controllable, and executable traffic demand for reliable evaluation, but real-world OD demand, routes, and turn movements are often missing or only sparsely observed.

Why labels are missing

- Limited sensing coverage means many links and turns are never directly observed.
- Privacy and data-sharing constraints prevent some traffic data from being publicly released.

Traffic Flow Generation

Generation target

- Realistic traffic flow files runnable by a simulator, consistent with real-world traffic patterns
- Controllable generation, allowing customization aligned with user preferences

Hierarchical generation:

OD-time $\rightarrow$ route $\rightarrow$ link/turn $\rightarrow$ flow

- OD-time demand decomposition
- Route assignment
- Link/turn flow projection
- Simulation flow-file generation

Main Contributions

- 1 We study traffic-flow generation for signal control algorithms.
- 2 We propose a hierarchical cascade from OD-time demand to routes, link/turn flows, and executable flow files.
- 3 We validate the framework on real networks and traffic counts across Austin, Chicago, NYC, San Francisco, and Washington DC under zero-shot, few-shot retrieval, and full-shot settings.

Place the method figure here as `Figures/Overview.png`.

Place the inference figure here as `Figures/inference.png`.